

Problem

Solve the integrodifferential equation

$$\frac{dy}{dt} + 4y + 3 \int_0^t y \, d\tau = 18e^{-2t}u(t), \quad y(0) = -3$$

Step-by-step solution

Step 1 of 4

The integrodifferential equation is,

$$\frac{dy}{dt} + 4y + 3 \int_0^t y \, dt = 18e^{-2t}u(t)$$

Apply the Laplace transform to the differential equation.

$$sY(s) - y(0) + 4Y(s) + 3 \left[\frac{Y(s)}{s} \right] = \frac{18}{s+2}$$

Substitute -3 for $y(0)$ in the above equation.

$$sY(s) - (-3) + 4Y(s) + 3 \left[\frac{Y(s)}{s} \right] = \frac{18}{s+2}$$

$$\left[s + 4 + \frac{3}{s} \right] Y(s) = \frac{18}{s+2} - 3$$

$$\left[\frac{s^2 + 4s + 3}{s} \right] Y(s) = \frac{18 - 3s - 6}{s+2}$$

$$Y(s) = \frac{12s - 3s^2}{(s^2 + 4s + 3)(s+2)}$$

$$= \frac{12s - 3s^2}{(s+1)(s+2)(s+3)}$$

Step 2 of 4

Apply partial fractions to above equation to simplify the equation.

$$Y(s) = \frac{A}{s+1} + \frac{B}{s+2} + \frac{C}{s+3} \dots\dots (1)$$

Calculate the constant A using residue method.

$$A = \left[(s+1)Y(s) \right]_{s=-1}$$

$$= \left[(s+1) \frac{12s - 3s^2}{(s+1)(s+2)(s+3)} \right]_{s=-1}$$

$$= \frac{12(-1) - 3(-1)^2}{(-1+2)(-1+3)}$$

$$= \frac{-15}{(1)(2)}$$

$$= -7.5$$

Step 3 of 4

Calculate the constant B using residue method.

$$\begin{aligned} B &= \left[(s+22)Y(s) \right]_{s=-2} \\ &= \left[(s+2) \frac{12s-3s^2}{(s+1)(s+2)(s+3)} \right]_{s=-2} \\ &= \frac{12(-2)-3(-2)^2}{(-2+1)(-2+3)} \\ &= \frac{-36}{(-1)(1)} \end{aligned}$$

$$= 36$$

Calculate the constant C using residue method.

$$\begin{aligned} C &= \left[(s+3)Y(s) \right]_{s=-3} \\ &= \left[(s+1) \frac{12s-3s^2}{(s+1)(s+2)(s+3)} \right]_{s=-3} \\ &= \frac{12(-3)-3(-3)^2}{(-3+1)(-3+2)} \\ &= \frac{-63}{(-2)(-1)} \end{aligned}$$

$$= -31.5$$

Step 4 of 4

Recall equation (1).

$$Y(s) = \frac{A}{s+1} + \frac{B}{s+2} + \frac{C}{s+3}$$

Substitute -7.5 for A , 36 for B and -31.5 for C in the above equation.

$$\begin{aligned} Y(s) &= \frac{-7.5}{s+1} + \frac{36}{s+2} + \frac{-31.5}{s+3} \\ &= -\frac{7.5}{s+1} + \frac{36}{s+2} - \frac{31.5}{s+3} \end{aligned}$$

It is known that,

$$L^{-1} \left\{ \frac{1}{s+a} \right\} = e^{-at}$$

Apply inverse Laplace transform.

$$\begin{aligned} y(t) &= -7.5e^{-t}u(t) + 36e^{-2t}u(t) - 31.5e^{-3t}u(t) \\ &= \left[-7.5e^{-t} + 36e^{-2t} - 31.5e^{-3t} \right] u(t) \end{aligned}$$

Therefore, the solution, $y(t)$ to the integrodifferential equation is

$$\boxed{\left[-7.5e^{-t} + 36e^{-2t} - 31.5e^{-3t} \right] u(t)}.$$